Historical research report

Round 3: Screening and priority proof packages

Review date: 4 November 2026 | Original report: 19 pages

Historical snapshot

This file preserves an earlier review. Later findings and the current selection supersede its recommendations, rankings and proposed next actions. Read the current shortlist and Round 4 before using this report for a development decision.

Current assessment

Current shortlist
github.com/KyleJKistner/ddiscov/blob/main/decisions/current-shortlist.md

Round 4 research and decisions
github.com/KyleJKistner/ddiscov/blob/main/research/rounds/round4/README.md

Original report preserved

All original report pages follow unchanged. The original report page numbers are retained; this archival cover is additional.

ddiscov | Screening round 3

More candidates, stronger comparisons, concrete next proofs

Research cut-off: 4 November 2026 | KyleJKistner/ddiscov | snapshot c241dc4

Decision
BLU3797 remains the first human-function proof. Add exact 8015-P2 and topical NDC-9009 to the front group. Keep AAT on a small formulation/potency gate. Stage DNM2-PPMO human translation and use FXT-001 as a serious organ benchmark. PLC-zeta moves down.

Work completed	Count and meaning
Additional candidate/use screens	36: 34 new to the reviewed inventory; 2 expansions of earlier concepts. These include negative screens and occupied programs.
Prior shortlist developed	6 of 6: BLU3797, AAT, CPTX, mIGF, FeM1269, PLC-zeta.
Research and challenge	Three domain teams plus integrating review; three independent cross-challenges.
Coverage retained	622 original coverage rows and 30 earlier proposals preserved. Neither is a count of unique drugs.
S-tier promotions	0. The review supports specific proof investments; it does not establish an obvious standalone company or clinical success.

What earns the next allocation

The priority is a result another group can reproduce: a defined article, a qualified established-disease deficit, useful function, a credible comparator and measured exposure. A small study that can overturn selection is more valuable than another encouraging marker or an undefined diligence conversation.

This report develops the four leading proof packages, identifies two secondary opportunities, updates the previous shortlist, and records all 36 additional screens. The accompanying cumulative JSON retains full evidence boundaries, source URLs, advance/stop conditions and reviewer provenance.

No experiments, sample receipt, outreach, purchases or remote repository changes are represented by this review. Rights and current development are recorded separately from scientific selection. No owner response is used as a gate.

Current proof priorities

Priority	Product and next decision	Why this position
1 | BLU3797	Established human MTM1 muscle force/fatigue rescue at measured exposure; genetic target concordance; neural/hepatic separation.	Strong target/animal rationale plus newly concrete independent model starting materials. First qualify chemistry, assay and model.
2 |8015-P2	Independent exact-stereoisomer rescue of mature human MFN2 neuromuscular function.	Strongest new post-onset animal lead. Human family evidence does not yet belong to pure P2.
3 |NDC-9009	Measured topical delivery followed by delayed Darier epidermal barrier/adhesion repair.	Compact disease-cell proof with clear exposure and active-comparator questions. Current result is junction assembly, not established-lesion reversal.
4 |AAT EVLP	Potency and composition-matched comparison before scarce paired-organ validation.	Human organ signal remains valuable. Excipient/protein effects and washout durability need an interpretable test.

Secondary staged opportunities

DNM2 Pip8b2-PMO6 merits a human-sequence translation package because late mouse force improves. It remains behind the core group while the actual human product and safety window are selected. FXT-001/UAMC-3203 has unusually relevant organ evidence, but exact human methods are incompletely inspected and the lead is already being developed by FiriX; a differentiated preservation experiment is needed.

Reserves remain specific

mIGF retains a practical formulation/comparator proof; FeM1269 retains a human iron-extraction-and-delivery proof; CPTX retains independent delayed-contusion replication. PLC-zeta drops to comparator-dependent reserve. Org43553, MG53 gel, XIB4035 and SAR247799 have narrower questions that could reopen priority.

How to read the order

This is an order for the next proof, not an estimate of approval probability. It combines relevance of the evidence, the ability to make the exact article, the interpretability of a functional test, and how much the answer would change the development decision. Differences in cost or model access can change execution order; no numerical finance score or unquoted budget is invented.

BLU3797: make human force the decision

The drug and independent genetic PIK3C2B work provide a coherent mouse rationale. This round found no new human drug-force result. It did narrow the chemistry claim: key final steps and analytical identity are public, while a complete independently verified synthesis is not established. The published live-cell selectivity panel also leaves C2alpha/PIK3CB comparisons unresolved. [S001] [S148]

The model route is more concrete

Coriell GM06813 is an MTM1 frameshift fibroblast background; GM24509 carries p.D394H. They supplement the Curi p.L463S route. A public patent describes deriving immature myotubes from the two Coriell lines. These are starting materials and a disclosed route, not three qualified force models or drug responders. [S004] [S005] [S006] [S149]

Addgene238834 supplies a public NanoLuc-PIK3C2B vector. It does not by itself reconstruct the study-specific tracer and engagement assay. [S003]

Proposed proof package

Stage1: authenticate the stereoisomer and relevant impurities; qualify an interpretable engagement measurement and an orthogonal genetic control. Establish a stable mature-muscle deficit and show that restoring MTM1 can rescue it. Otherwise a negative drug result would not be interpretable.

Stage2: randomize after that deficit exists. Measure force, fatigue and excitation-contraction performance across independent backgrounds, with correction/restoration controls. Quantify free medium and tissue exposure alongside engagement. More myotubes or a prettier structural image does not answer this decision.

Stage3: compare the effective free-exposure range with neural-network and hepatobiliary function. Constitutive human target genetics identifies a reason to examine neural risk; it does not give a drug concentration threshold. Distinguish failure of this molecule from failure of the target.

Advance / stop
Advance on reproducible target-concordant human functional recovery with a usable neural/hepatic margin. Stop the article if engaged inhibition fails in a genetically rescuable model or useful exposure conflicts with healthy-tissue function. A peripheral chemistry project would then be a separately scored proposal.

Transferable output: authenticated material specification, model qualification data, exposure-response curves, source-level genotype provenance and a prespecified functional go/no-go conclusion.

8015-P2: the strongest new reversal lead

The 2026 rat study starts after about 50 weeks of established disease and reports randomized, blinded functional assessment. Dose cohorts are small. Reported normalization partly relies on nonsignificance versus WT, which is not equivalence. Acute motility measurements at 100 mg/kg do not establish engagement at the lower chronic doses. [S010]

The 2024 work specifies chiral 8015-P2 and shows functional reversal in knock-in mice. Its human T105M fibroblast/neuron experiments use the 8015 mixture from one patient. Earlier four-patient work uses Chimera C. These cannot be pooled into an exact-P2 human efficacy claim. The animal studies substantially overlap investigators and the T105M genotype; two species do not supply independent replication. [S011] [S012]

Material and human-model route

The public synthesis/chiral separation supports a defined custom-material specification. Require peak identity, stereochemistry, purity and biological activity; the company-supplied material in a paper is not a received research lot.

The NYSCF/CMTA catalog names MFN2 H361Y, R364W and c.2219G> C iPSC lines and fibroblast availability. Its distribution is request-based, and catalog dates are inconsistent with the URL date. This resolves a named repository route, not current stock or qualified neurons. A public T105M patient stock was not verified. [S056] [S057]

Proposed proof package

Use independently authenticated pure P2, parent mixture, less-active P1 and vehicle in mature human neuromuscular tissue after a stable deficit. Start with a reconstructable genotype and add a distinct pathogenic background; include isogenic restoration/correction. Catalog patient material and engineered lines answer related, but not identical, questions.

Measure nerve-evoked force, transmission reliability and fatigue, then directly stimulate muscle to separate neuronal and muscular contributions. Mitochondrial transport and morphology explain mechanism; they do not replace integrated function. Compare pulsed and continuous exposure and quantify actual unbound exposure at the tissue.

Assess healthy neuronal and cardiac function at the same exposure. A limited selectivity panel divided by an assay EC50 is not a therapeutic index. Check whether altered excitability, cell growth or mitochondrial quality-control impairment could produce a misleading short-term benefit.

Advance / stop
Advance exact P2 only for reproducible established human functional recovery with an exposure window. Stop or narrow if only the mixture works, benefit is confined to one genotype, or apparent improvement is generalized hyperexcitability. A T105M-specific result would support a narrower claim, not all CMT2A.

Existing mitofusin patent-family claims require separate value-capture analysis. A ceased PCT entry does not erase a granted US family member. [S014]

NDC-9009: link topical delivery to repair

The patient-cell adhesion experiment uses 5 micromolar NDC-9009 during junction assembly. Four isolates represent three donors and two ATP2A2 variants. It is a direct mechanical readout, but it does not establish delayed repair in mature skin. CDN1163 supplies some separate mechanistic results. [S098]

The chemistry paper provides a synthesis and reports PBS pH 7.4 solubility of 0.23 micromolar. The nominal Darier concentration is about 22 times that value. Culture cosolvent/protein and skin partitioning differ; this does not prove invalid activity or an absolute skin-exposure ceiling. [S099]

A specific failure mode and a stronger comparator

NAC protects mechanical adhesion in the same disease model. The authors also propose that stress can inactivate residual SERCA. An activator could therefore protect junction formation yet fail after an established lesion has lost the activatable protein pool. [S098]

Proposed proof package

First authenticate NDC-9009 and measure dissolved and free exposure in the assay formulation. Then measure delivery into viable epidermis after topical application. Surface residue and total whole-skin concentration cannot establish exposure of the relevant living cells. A vehicle that changes cohesion or irritation must be separated from drug action.

Establish a persistent adhesion/permeability deficit in mature epidermis before randomization. Include a preventive arm to distinguish flare prevention from lesion reversal. Compare vehicle, NAC and CDN1163 at relevant achieved exposures; add a clinically relevant treatment when the model can measure its mechanism and benefit. Do not infer superiority from micromolar versus millimolar labels.

Use independent genetic backgrounds, measure mechanical cohesion and barrier recovery, and challenge durability after washout or renewed stress. Healthy tissue should retain barrier, differentiation and structural integrity. Viability alone is insufficient to define a skin window.

A public engineered SERCA2 keratinocyte model supports an independent reconstruction route. The original model and a newly engineered catalog-cell substitute must be qualified separately. The ATCC Darier listing is an archival fibroblast Mission Collection item without a public ATP2A2 genotype; it is not a ready patient-epidermis model. [S102] [S103]

Advance / stop
Advance for delayed functional repair at achieved epidermal exposure with a healthy-skin margin and a meaningful advantage over simpler treatment. Stop or narrow if only prophylaxis works, formulation explains the result, or a simpler comparator matches the useful benefit. The oral dantrolene trial results remain unverified; completion is not efficacy.

AAT: isolate biological activity from the formulation

The 2026 human report describes eight donor blocks with paired lungs. Earlier reporting should not be added as an independent cohort. Full current methods were not recovered, so exact human dose and numerical magnitude are not imported from pig work or the patent pilot. [S036] [S037]

The patent pilot uses 10 mg/mL AAT against saline. Zemaira labeling specifies approximately 50 mg/mL reconstitution plus 121-168 mM mannitol and 15-20 mM phosphate, and administration within 3 hours. The label does not validate prolonged warm perfusion. [S038] [S039]

Calculated formulation scenario
If a 50 mg/mL stock supplies 10 mg/mL final AAT, it contributes 20% of final volume: about 24-34 mM mannitol and 3-4 mM phosphate. This is a calculation for a candidate recipe, not a reconstruction of the 2026 human experiment or proof that its signal is confounded.

Proposed proof package

Begin with a low-burden perfusate qualification: functional elastase-neutralizing potency, aggregation, loss to the circuit and stability over the intended warm dwell. Record active potency as well as nominal protein mass. Match pH, osmolality, ionic composition, protein/oncotic load and relevant excipients in the comparator.

A biologically inactive protein control needs independent qualification; heat treatment can change aggregates and physical properties. Use the minimum set of controls that separates activity from the rival physical explanation, rather than interpreting any inactive preparation as a perfect match.

Move to paired donor lungs only after the formulation contrast is interpretable. Analyze the donor as the biological block, with oxygenation, compliance and fluid handling as the decision endpoints. Molecular inflammation markers explain the result, but cannot substitute for useful graft physiology.

Then challenge the benefit after removing the drug and during recipient-blood reperfusion. A durable ex vivo-only claim ultimately needs transplant validation with measured recipient carryover. Systemic recipient dosing is a different product and cannot supply that missing evidence.

Advance / stop
Advance if active AAT adds meaningful donor-level function after physical matching and the benefit persists through a relevant washout challenge. Stop or redesign if matching explains the signal, activity decays in the circuit, or improved perfusion physiology does not survive subsequent reperfusion.

The first investment is therefore smaller than a full donor-organ efficacy campaign. It can supply a valuable product specification or terminate the thesis before consuming scarce material.

STAGED DEVELOPMENT

Two serious secondary opportunities

DNM2 Pip8b2-PMO6: select the human product before borrowing efficacy

Late treatment in Dnm2 R369W mice improved specific muscle force from 11.2 to 15.6 mN/mg, with four treated versus five disease-control animals. Readout was four weeks after the last dose. Protein fell, but strong nonsense-mediated RNA decay was not demonstrated. The mouse construct is not the separately patented human oligo. [S015] [S006]

The proposed first package selects a publicly specified human splice oligo, authenticates the conjugate, and confirms its splice/protein consequence. In mature DNM2-mutant muscle, compare post-deficit force/fatigue with a non-targeting conjugate, unconjugated PMO and orthogonal DNM2 reduction at comparable protein suppression. Examine muscle benefit alongside kidney, liver, cardiac and innate-immune effects. Absence of short-term mouse toxicity does not establish a human chronic window.

Advance only if the actual human construct reproduces useful function at tolerable exposure. Stop if target reduction is measurable but function fails, if an unintended product persists, or if useful muscle exposure overlaps non-muscle injury. This is a worthwhile translation project with an additional product-selection step.

FXT-001/UAMC-3203: strong preservation evidence, occupied program

The 2026 Cell abstract reports porcine liver/lung protection and preservation of declined human lungs during perfusion. Full primary methods were not extracted from the large author manuscript. Human donor count, dose, effect magnitude and exact salt/formulation remain unverified in this review. [S058] [S068]

Public chemistry and research specifications identify UAMC-3203; free-base and hydrochloride forms differ. The university identifies FXT-001 as FiriX Therapeutics' lead. This is an active external program. [S061] [S069]

The differentiated proof would establish an exact short-exposure graft-conditioning specification: solubility, protein binding, circuit loss, tissue retention and postwashout function against optimized perfusate and a credible additive benchmark. Preservation is a legitimate objective. A delayed repair claim requires treatment after a documented deficit and actual reversal; it cannot be inferred from deterioration prevented.

Allocation decision
Keep DNM2-PPMO as a serious human-translation reserve. Keep FXT as an organ benchmark and a conditional differentiated proof, with full-method reconciliation before scarce-organ allocation. Neither gains priority merely because it is newly found.

The previous shortlist: material changes

mIGF corneal gel: retain; qualify the article and comparator

Figure8 of the primary paper reports corneal-sensitivity improvement in the steroid-delayed model, n=3. The paper includes single-motif, scrambled and blank-gel controls. Its205 micromolar aggregation threshold, 0.4 wt% gel and nanomolar cell assays leave the released active species/exposure unresolved. [S045]

First qualify two independent peptide lots for assembly, release, potency and optical clarity. Then use paired corneal tissue under washout against validated topical insulin, free IGF-1 and matched vehicle. Require durable barrier/adhesion with a meaningful dosing or functional advantage. Sensation requires a separate innervated model; an ex vivo cornea cannot answer that endpoint. The sensory result improves the case, but its small size and comparator gap keep it conditional.

CPTX: retain independent delayed-injury replication

The exact-protein primary route is local intraspinal injection. A separate 2024 CPTX-expressing-cell study is a different article. The current search did not verify an independent exact-protein delayed-contusion replication; this is a bounded search result, not proof none exists. [S052] [S053]

Qualify protein distribution and the window between useful evoked transmission and pathological activation before a large behavioral study. Then test delayed incomplete contusion with baseline stratification, blinded coordinated movement/conduction and matched rehabilitation. Changed route, tag or carrier requires comparability. No new paper count supplies the missing independent functional proof.

FeM1269: isolate extraction from iron delivery

Hinokitiol work supplies three patient macrophage backgrounds, but not exact-FeM efficacy. A separate human erythroid-culture paper shows useful iron-loaded-deferiprone delivery, hemoglobinization and oxygen-binding behavior. That process input is not an anemia treatment or proof of extraction from iron-restricted macrophages. [S047] [S048]

Track macrophage-derived iron into erythroid heme under persistent restriction. Use holo-transferrin as a positive control and iron-loaded deferiprone as a delivery benchmark; keep loaded and unloaded chelators distinct. Require useful maturation without labile-iron misdistribution. Delivery alone cannot validate the inflammatory-anemia thesis.

PLC-zeta: move down until developmental advantage exists

A 2020 direct comparison in PLC-zeta-null sperm models reported rhPLC-zeta versus ionomycin blastocyst rates of 41% versus 50%; superiority was not established. The 326 oocytes were distributed across conditions, not 326 donors. This does not prove inferiority or equivalence. [S041]

Physiological calcium oscillations are insufficient product differentiation. An exact reagent must produce normal developmental or handling advantage against ionomycin in the intended subgroup, with donor-level analysis. Retain it as a comparator-dependent reserve rather than a front-line financing thesis.

Current evidence that prevents wasted work

POLG: sulbactam has a stronger benchmark

PZL-A has patient-fibroblast replication/respiratory evidence in the 2025 Nature study. Pretzel' s 30 October 2026 statement reports regulatory authorization and planned patient development for the separate asset PX578. PZL-A and PX578 must not be equated; planned trial timing is not patient efficacy. [S144] [S145]

Sulbactam should earn a feasible-exposure, functional or genotype advantage in endogenous patient systems against authenticated PZL-A. An engineered aggregation readout alone is now a weak development proposition. This is an expanded prior comparator, not a newly invented POLG family.

BRD4780: clearance can conceal broad effects

The independent September2026 preprint reproduces MUC1-fs lowering while reporting wild-type MUC1 suppression and broad transcriptional effects. This is preliminary, unreviewed counterevidence; it is not a definitive clinical failure. [S120]

Attach the selectivity question to every old positive summary. A useful follow-up tests normal secretory function, polarity, transport and viability alongside disease function, not another mutant-clearance image.

Human negatives should travel with the mechanism

DRCR' s 129-participant tonabersat trial did not meet its primary DME thickness endpoint. NMD' s 81-participant CMT phase 2a release likewise reports a missed primary 6-minute-walk endpoint, despite positive secondary measures. Different mechanisms, genotypes and treatment durations must remain distinct. [S150] [S026]

These findings stop promotion based solely on older leak, inflammation or excitability stories. They do not establish that every drug acting in either disease will fail. A new candidate needs a relevant, discriminating functional claim.

Occupied or incomplete programs are not automatically failed

Ripasudil, mirococept, M101, surlorian and KUS121 belong in the current comparator set. The alphaCT1 phase 3 termination record does not justify an efficacy-failure label. Dantrolene' s Darier results document is listed, but actual outcomes and actual enrollment were not recovered. Source URLs and specific boundaries accompany each screen below.

Implication
A lead survives when it can answer a better question than the current program or comparator. Known development does not prohibit a valuable new proof, but it defeats an unsupported claim that the article is neglected or commercially open.

Neuromuscular and metabolic screens

Each row is an article/use assessment. Novel to this register does not mean a new invention, unowned asset, positive result or development-ready lead. Full donor/exposure/material/rights boundaries and go/stop conditions are in the cumulative register.

Article/use and disposition	Decision-changing evidence / sources	Next discriminating question
R3-NM-01 8015-P2 / CMT2A CORE PROOF	Established animal functional reversal. Exact human P2 result still missing; mixture/Chimera C are separate articles. [S010] [S011] [S012] [S013] [S014] [S056] [S057]	Independent mature human neuromuscular function; measured exposure and nerve-versus-muscle separation.
R3-NM-02 Pip8b2-PMO6 / DNM2-CNM STAGED PROOF	Late mouse force rescue persists to a month after dosing. Human sequence/conjugate selection remains. [S015] [S006]	Actual human oligo must rescue established force with a non-muscle safety window.
R3-NM-03 CKD504 / CMT2D HOLD	Neuronal maturation and WT firing changes confound the patient signal; exact code-to-structure mapping unresolved. [S016] [S017] [S018]	An identified article must rescue established useful function without generalized excitability.
R3-NM-04 XIB4035 / small-fiber dysfunction RESERVE	Older post-onset sensory improvement is useful; catalog identifiers conflict. [S019] [S020] [S021] [S022]	Resolve article identity; distinguish sensory restoration from altered reflex threshold.
R3-NM-05 NV354 / complex-I Leigh DEPRIORITIZE	Predominantly prevention, no lifespan extension, and parent/metabolite exposure uncertainty. [S023]	Only reopen for a qualified post-onset energy-demand function question.
R3-NM-06 Surlorian / RYR1 BENCHMARK	Small open-label human evidence; active phase 2 program. Reenrollment is not another independent patient. [S024] [S025]	Require a distinct genotype, route or response-selection need before duplicate development.
R3-NM-07 Ignaseclant / CMT BENCHMARK	Sponsor-reported randomized study missed primary walking endpoint; secondary outcomes favored treatment. [S026] [S027]	Carry the primary negative into endpoint and differentiation logic; do not generalize to all CMT.
R3-NM-08 Decitabine / recessive RYR1 HOLD	Mouse muscle function does not establish chronic separation from marrow/DNA effects. [S028] [S029] [S030]	Measure mature patient-force rescue against matched marrow and DNA-injury exposure.
R3-NM-09 ARD-1676 / SBMA MECHANISTIC RESERVE	Mutant-AR and transcript effects are not force, motor or survival recovery. [S031] [S032] [S033]	Require established human neuromuscular function, normal AR preservation and exposure.

Organ and blood screens

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Article/use and disposition	Decision-changing evidence / sources	Next discriminating question
R3-OR-01 FXT-001 / organ preservation CONDITIONAL / BENCHMARK	Primary abstract supports direct human-lung preservation; full human methods incomplete; active FiriX lead. [S058] [S059] [S060] [S061] [S062] [S063] [S064] [S065] [S066] [S067]	Qualify exact form/exposure, then show differentiated sustained function beyond optimized perfusate.
R3-OR-02 CYM5442 / donor lung COMPARATOR	Seven donor pairs: leak/pressure signal without significant compliance, P/F or PVR improvement; S1PR1 falls. [S070] [S071] [S072]	Test useful organ function and receptor recovery after withdrawal.
R3-OR-03 SAR247799 / pulmonary barrier SUCCESSOR RESERVE	Bias may address desensitization, but sponsor terminated development and microvascular evidence was unconvincing. [S073] [S074] [S075] [S076] [S077] [S078] [S079]	Head-to-head pulmonary barrier/washout proof versus CYM before whole organs.
R3-OR-04 PTL060 / thrombotic graft injury NARROW RESERVE	Human feasibility is not filtration rescue; primate histological benefit did not improve survival. [S080] [S081] [S082]	Postwashout graft function against free thrombin inhibition and inactive anchored control.
R3-OR-05 MitoQ / kidney preservation DEPRIORITIZE	Flow signal coexists with independent renal mitochondrial-injury evidence. [S083] [S084]	Resolve exposure-toxicity and demonstrate transport/filtration, not flow alone.
R3-OR-06 APT102 / AZD3366 RESERVE	Canine reperfusion function and human PK/PD exist; ex vivo-only benefit is a new route hypothesis. [S085] [S086] [S087]	Retained graft benefit with bounded residual enzyme/antiplatelet action after washout.
R3-OR-07 M101 / HEMO2life PRODUCT BENCHMARK	Prior human paired-organ work and marketed-device context; definitive OXYOP2 results not verified. [S088] [S089] [S090]	Require a specific handling or functional advantage over the existing product.
R3-OR-08 Mirococept / APT070 ACTIVE BENCHMARK	Initial low-dose study lacked DGF benefit; higher organ-directed dosing is under study. [S091] [S092]	Dose by target burden/retention; do not restart a generic complement thesis.
R3-OR-09 Sodium thiosulfate / kidney ACTIVE BENCHMARK	Exact preservation use already in a recruiting trial; new pig evidence is abstract-level. [S093] [S094] [S095]	A separate proposal must identify an unanswered exposure, timing or functional advantage.
R3-OR-10 Sphingosine / donor-lung decontamination RESERVE	Pig antibacterial effect; human explants show delivery, not replicated antibacterial efficacy. [S096] [S097]	Viable clearance beyond antibiotics while preserving clinically representative graft function.

Local and rare-disease screens

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Article/use and disposition	Decision-changing evidence / sources	Next discriminating question
R3-NL-01 NDC-9009 / Darier CORE PROOF	Direct patient-cell adhesion signal; three donors; assembly timing; solubility/exposure unresolved. [S098] [S099] [S100] [S101] [S102] [S103]	Measured viable-epidermis delivery, delayed repair and advantage over NAC/active treatment.
R3-NL-02 Dantrolene / Darier COMPARATOR / RESERVE	Public oral-trial completion and results-document listing do not reveal actual outcomes. [S104] [S105] [S106] [S107] [S108]	Read actual result if accessible; separately test whether local exposure earns an advantage.
R3-NL-03 Trametinib / Darier REFINEMENT	Existing drug in inventory, new local-use comparison. Induced-model protection is not patient-lesion repair. [S102]	Delayed mature-skin function and healthy-epidermis margin against calcium-restoring comparators.
R3-NL-04 Org43553 / selected LHCGR RESERVE	Adult mutant cycles/ovulation indicators improve; treated-mutant live-birth rescue absent. Human exposure spans two compounds. [S109] [S110] [S111] [S112] [S113] [S114]	Exact salt/genotype, attainable exposure and reproductive competence beyond trafficking/cycles.
R3-NL-05 MG53 ophthalmic gel RESERVE	Acute injury optical/vascular outcomes improve; both groups close byday 5; n descriptions conflict. [S115] [S116] [S117]	Delayed optical/barrier recovery beyond blank gel, solution and practical corneal repair.
R3-NL-06 BRD4780 / ADTKD-MUC1 DOWNRANK	Independent preprint confirms mutant lowering but challenges selectivity. [S118] [S119] [S120] [S121]	Normal cargo handling and renal-cell function must improve alongside mutant clearance.
R3-NL-07 MANF protein / ADTKD-UMOD DELIVERY RESERVE	Organoid evidence uses induced gene expression, not the proposed exogenous protein. [S122] [S123]	Demonstrate protein compartment delivery and useful transport function after deficit.
R3-NL-08 AlphaCT1 / chronic wounds HUMAN BENCHMARK	Human wound evidence and current development exist. Phase3 termination is not a verified efficacy failure. [S124] [S125] [S126] [S127]	Any distinct product must improve durable closure, barrier or recurrence beyond current care.
R3-NL-09 KUS121 / retinal ischemia CLINICAL BENCHMARK	Small uncontrolled human study; later randomized program exists but numerical result not recovered. [S128] [S129] [S130] [S131]	Resolve sham-controlled outcomes before a duplicate proposition.
R3-NL-10 Ripasudil / FECD after DSO STRONG BENCHMARK	Randomized human corneal-edema recovery and ongoing phase 3. [S132] [S133]	A new product needs a meaningful unresolved patient or treatment-burden advantage.

Article/use and disposition	Decision-changing evidence / sources	Next discriminating question
R3-NL-11 Tonabersat / good-vision DME REJECT BROAD LEAD	Randomized primary endpoint not met; exploratory subgroup does not rescue the general thesis. [S134]	Only a distinct, prespecified independently justified responder hypothesis could reopen.
R3-NL-12 ML141 / ADPKD PROBE RESERVE	Human organoid cyst shrinkage is reversible and does not show restored renal function. [S135]	Direct transport/secretion, viability and achievable exposure against tolvaptan.
R3-NL-13 Amlexanox / PAX6 nonsense EARLY RESERVE	Genotype-relevant developmental models; no established adult barrier/vision recovery. [S136] [S137] [S138]	Mature variant-matched corneal function and full-length functional PAX6 at local exposure.
R3-NL-14 rhTFF3 / corneal repair LOW PRIORITY	Defined article with acute repair data; weak chronic-defect/comparator differentiation. [S139] [S140]	Delayed persistent-defect barrier and clarity beyond active corneal repair.
R3-NL-15 VP1-001 / selected cataract CONFLICTED RESERVE	Optical-surrogate signals; independent human-lens findings differ by article/assay. [S141] [S142] [S143]	Blinded exact-article human-lens optics with measured intralenticular exposure.

Two comparator/manufacturing extensions

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Article/use and disposition	Decision-changing evidence / sources	Next discriminating question
R3-RT01 Iron-loaded deferiprone / RBC manufacture PROCESS BENCHMARK	Human cultured red-cell function supports an ex vivo input, not clinical anemia treatment. [S048] [S049]	Scale, yield, quality, residuals and all-in process cost must improve.
R3-RT02 PZL-A / POLG comparator EXPANDED COMPARATOR	Already named in prior POLG proposal. Exact patient-cell evidence and a separate active clinical program raise the bar. [S144] [S145] [S146] [S147]	Sulbactam must show added endogenous patient function or a complementary niche.

Make the next result change a decision

1. Separate an article failure from a model failure

Establish a stable functional deficit and a credible rescue control before testing the lead. If the model cannot recover under its positive control, a negative compound experiment does not reject the mechanism. Conversely, engagement without useful function in a qualified model is a meaningful negative. Publish that distinction in the development package.

2. Use independent biological units

Donor, animal or patient background is the unit of generalization. Wells, neurites, sections, time points and repeated differentiations are nested measurements. Use donor-blocked allocation, blinding and hierarchical analysis where appropriate. Additional clones improve measurement but do not create more patients. Determine variance and a meaningful functional threshold before the efficacy comparison is unblinded.

3. Keep the rival explanation in the comparator

For AAT, it is perfusate composition. For NDC, it is formulation, junction assembly and a simpler antioxidant. For P2, it is parent-mixture activity or direct muscle/excitability effects. For FeM, it is supplying iron without extracting restricted stores. For PLC-zeta, it is equivalent development with a simpler activation reagent. The comparator should answer the actual reason the result might fail to create value.

4. Treat exposure and timing as part of the product

A compound name does not specify free target-tissue exposure. A preventive co-treatment does not prove established-disease reversal. A changed route, stereoisomer, sequence, salt or formulation is a separate comparability task. These rules let an independently made material produce an interpretable result without needing privileged investigator assurances.

5. Distinguish a fundable proof from a fundable company

The immediate value proposition is a bounded result that would make a developer change its decision. An independent company additionally requires a credible product advantage and value-capture route. Public patents and active programs can narrow that route without invalidating the biology. None of the selected leads is called commercially open or obviously financeable on the present public record.

Specific remaining limits

No new wet-lab data, received material, current physical inventory or institutional donor-organ access was generated. FXT human full-method extraction and exact test form remain incomplete. AAT 2026 full methods were not recovered. Dantrolene Darier and KUS121 later-trial numerical results remain unverified. mIGF supplementary-method details remain incomplete. These uncertainties are attached to the records; they do not become positive evidence or a request to contact an owner.

Search novelty is relative to the reviewed repository and prior register. The 34 additions include existing clinical programs and negative screens. The six previous-shortlist updates are not six new discoveries. This focused extension is substantial but not an exhaustive worldwide systematic review.

Source index

Source labels are stable within this report and the round 3 section of the register. Click any reference or URL. Primary research supports biological claims; issuer/registry pages support their own program status; catalogs support identity/listing only. Patent displays are not a claim-by-claim legal opinion. Raw team references and retrieval limits are preserved in the register.

S001 BLU3797 primary study and synthesis
https://insight.jci.org/articles/view/198568

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S003 NanoLuc-PIK3C2B fusion vector 238834
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S004 Coriell MTM1 fibroblast GM06813
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S006 Antisense oligomers for treatment of centronuclear myopathies
https://patents.google.com/patent/WO2025085810A2/en

S007 PIK3C2B variants and focal epilepsy
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S008 PI3KC2 inhibitor medicinal chemistry; PI-701 limitations
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S013 Mitofusin activator orphan designation
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https://rycarma.com/news/rycarma-announces-first-patient-dosed-in-phase-2-clinical-trial-of-surlorian-for-the-treatment-of-ryr1-related-myopathy/

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S027 NMD Pharma current releases
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S028 Low-dose decitabine in recessive RYR1 myopathy mice
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